

Statement of Contribution

1. General Role in the Project

My responsibilities included collecting relevant research papers to support the literature review and conducting a detailed analysis of the v dataset. This involved data cleaning, processing, and extracting key insights related to pollution patterns. I also took the initiative to thoroughly understand all aspects of the project, including the contributions of my team members. This holistic understanding enabled me to contribute meaningfully to the writing of the results and discussion sections of the IEEE report.

2. Technical Contribution

- Collected and reviewed previous research papers related to urban air quality analysis to inform the study's context and direction.
- Loaded and preprocessed the `chicago_eclipse_data_part.csv` dataset (comprising five parts) using the Pandas library.
- Converted time formats and implemented techniques to handle missing values.
- Applied H3 hexagonal spatial encoding to aggregate the Chicago data geographically.
- Performed exploratory data analysis (EDA), including generating histograms for key variables such as PM2.5, temperature, and humidity.
- Implemented KMeans clustering on the H3-encoded data to identify spatial pollution patterns across Chicago.
- Exported final outputs to `chicago_summary_stats.csv` and `chicago_aggregated_h3.csv` for inclusion in the results.
- Assisted team members by validating outputs from other cities and ensuring code consistency and standardization across the entire project.

"I have attached screenshots of my code to clearly demonstrate my hands-on work in reference section "

Project Title: Comparative Spatial Analysis of Urban Air Quality: NYC, Boston, and Chicago

3. IEEE Paper Contribution

I contributed to writing and editing the results and discussion sections of the IEEE paper, with a focus on data interpretation and policy insights. This includes:

- **Comparative analysis of air quality across cities**, where I helped interpret cross-city differences in pollutant levels and identified statistically significant trends.
- **Exploration of PM2.5 pollution hotspots**, using spatial clustering results to highlight the most affected zones in Chicago, NYC, and Boston.
- **Analysis of environmental influences**, such as the role of temperature and humidity in shaping pollution patterns across different urban climates.
- **Discussion of policy implications**, where I outlined how localized pollution zones and seasonal variations support the need for city-specific regulations.
- **Drafting key policy recommendations**, including targeted interventions for traffic-heavy corridors, integration of weather data into forecasting systems, and strategic monitoring enhancements.

"Please refer to the green-highlighted sections in the attached IEEE paper in reference section"

4. GitHub Pull Requests (PRs) & Repo Structure & Organization

I contributed extensively to the main Jupiter Notebook file titled `NYC_Boston_Chicago_Project.ipynb`, where I implemented and documented the complete analysis pipeline for the **Chicago** dataset. This included data loading, preprocessing, H3 spatial encoding, clustering, regression modeling, and visualization of results.

Our team followed a **shared-editing workflow**, contributing directly to the main branch instead of using individual pull requests. Within this setup, I actively helped manage the GitHub repository structure to support smooth team collaboration. Specifically:

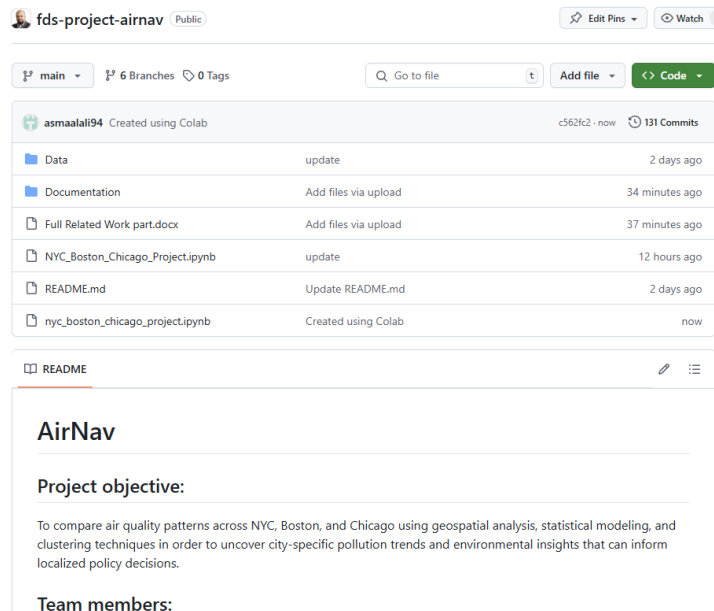
- I uploaded the **Chicago datasets** to the `Data/` directory.
- I contributed to the `Documentation/` folder by adding algorithm explanations, relevant formulas, and references to prior research that supported our methodology.

These contributions ensured clarity, reproducibility, and well-organized project documentation for all team members.

Name: Asma Abdalla Al Ali

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Project Title: Comparative Spatial Analysis of Urban Air Quality: NYC, Boston, and Chicago



6. Peer Review & Team Involvement

Throughout the project, I maintained active collaboration with my teammates by consistently reviewing their work, providing constructive feedback, and ensuring alignment with our overall objectives. I regularly participated in team discussions, supported others in refining their contributions, and helped maintain a cohesive and professionally structured final output. Additionally, I contributed to organizing and managing the shared project files, streamlining collaboration and improving workflow efficiency for the entire team.

Declaration

I confirm this document truthfully represents my contribution to the group project.

Signature:

A handwritten signature in black ink, appearing to be 'Asma', written over a horizontal line.

Name: Asma Abdalla Al Ali

Email ID: U24200616@sharjah.ac.ae

Project Title: Comparative Spatial Analysis of Urban Air Quality: NYC, Boston, and Chicago

Reference:

A Comparative Spatial Analysis of Urban Air Quality: New York City, Boston, and Chicago

Wafa Alyammahi¹, Mera Alshamsi², Asma Al Ali³, Nawaf Alnaqbi⁴, Isam Al Jawarneh⁵

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across all iterations, we evaluated the variance and confidence intervals of the β -values. This told us something about the stability of the temperature and humidity effects on PM2.5, especially with data noise or regional heterogeneity included. Bootstrapping also made sure that the inferences from the model did not rely on a specific sample split.

Root Mean Square Error (RMSE) was computed to test the predictive ability of the model. RMSE computes the mean absolute size of the prediction error and is extremely sensitive to large errors and thus is very helpful to trap poor fits.

The formula used is:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (4)$$

A lower RMSE value indicates better predictive performance. Together, bootstrapping and RMSE ensured both the statistical reliability and forecasting precision of the model, validating its use for air quality interpretation across the studied cities.

VI. RESULTS AND DISCUSSION

A. Comparative Analysis of Air Quality Across Cities

Our analysis revealed distinct air quality patterns across the three metropolitan areas. Chicago exhibited the highest average PM2.5 concentration at 16.8 $\mu\text{g}/\text{m}^3$, exceeding both New York City (13.2 $\mu\text{g}/\text{m}^3$) and Boston (9.7 $\mu\text{g}/\text{m}^3$). These differences were statistically significant ($p < 0.001$), suggesting fundamental variations in pollution sources and urban dynamics. PM10 data, available only for New York City, showed considerable spatial variation with higher concentrations in Manhattan and parts of Brooklyn, likely due to traffic density and construction activities.

Cluster analysis identified three distinct pollution profiles in each city: low, moderate, and high exposure zones. The high-exposure cluster in Chicago encompassed 28% of the analyzed area, compared to 21% in New York and only 15% in Boston. Interestingly, these high-exposure clusters were not uniformly distributed but showed specific geographic patterns—concentrating along major transportation corridors and industrial zones in all three cities.

The regression analysis demonstrated that meteorological factors explained a significant portion of PM2.5 variability, with R^2 values of 0.43, 0.38, and 0.51 for New York, Boston, and Chicago respectively. Temperature exhibited a positive correlation with PM2.5 levels in all cities, with a stronger effect in summer months. This may be attributed to increased photochemical reactions in warmer conditions, which accelerate secondary aerosol formation. Humidity showed a more complex relationship, with negative correlations during winter months but positive associations during summer, possibly reflecting seasonal differences in pollution formation mechanisms.

B. Policy Implications

These findings have several implications for urban air quality management. First, the significant inter-city differences suggest that uniform national regulatory approaches may be insufficient—targeted, city-specific strategies are necessary. The identified high-pollution clusters represent priority zones where enhanced monitoring and mitigation measures could yield the greatest public health benefits. The strong influence of meteorological factors on pollution levels indicates that air quality forecasting and warning systems should incorporate weather predictions to improve accuracy.

Furthermore, the spatial consistency of pollution hotspots along transportation corridors across all three cities suggests that traffic management and vehicle emission reduction strategies could have substantial benefits. The seasonal variations observed in the pollution-weather relationship also indicate that different control strategies may be optimal during different times of the year.

C. Air Quality Differences Between NYC, Boston, and Chicago

The bar chart comparison of mean PM2.5 concentrations across the three cities shows significant differences. Chicago has the highest PM2.5 levels (12.75 $\mu\text{g}/\text{m}^3$), followed by NYC (6.90 $\mu\text{g}/\text{m}^3$), with Boston having the lowest levels (1.84 $\mu\text{g}/\text{m}^3$). The variability also differs substantially - Chicago shows both high mean values and high standard deviation (12.06), suggesting more extreme pollution events. NYC similarly shows high variability (std=13.99), while Boston's air quality is both better and more consistent (std=0.77).

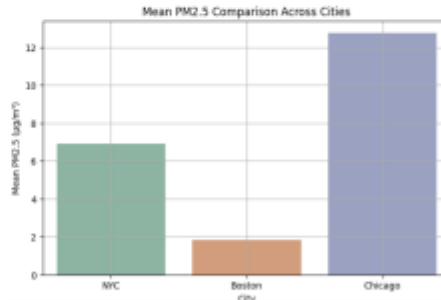


Fig. 1: Mean PM2.5 Comparison across cities

city	mean	std	min	max
0 NYC	6.896571	13.985542	0.000000	384.130000
2 Boston	1.837699	0.765489	0.464346	8.028841
4 Chicago	12.746455	12.057844	0.000000	2010.439943

TABLE II: Statistical data for different cities

This significant variation (nearly 7x difference between Boston and Chicago) suggests that regional factors, industrial

activity, population density, and traffic patterns play crucial roles in determining urban air quality.

D. Highest PM2.5 Pollution Areas

H3 geospatial indexing was used to identify pollution hotspots within each city. The clustering results, showing PM2.5 vs. other variables (PM10 for NYC, NO2 for Boston), help identify areas with similar pollution characteristics.

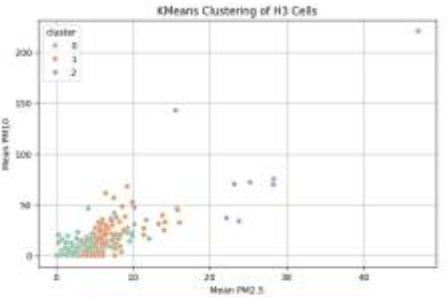


Fig. 2: Clustering analysis for all cities

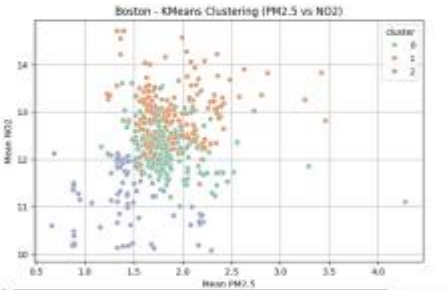


Fig. 3: Boston - KMeans Clustering (PM2.5 vs NO2)

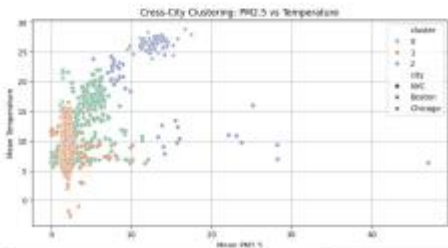


Fig. 4: Cross-City Clustering: PM2.5 vs Temperature

The clustering analyses reveal that each city has distinct pollution zones. In NYC, there's a cluster with particularly high PM2.5 and PM10 values (cluster 2 with PM2.5 mean of 26.69 $\mu\text{g}/\text{m}^3$ and PM10 mean of 90.18 $\mu\text{g}/\text{m}^3$). These hotspots likely correspond to areas with high traffic congestion or industrial activity.

E. Patterns Related to Weather and Urban Mobility

The clustering analysis reveals clear relationships between air pollution and meteorological conditions. Looking at the "Cluster Profiles" for NYC, cluster 2 shows not only the highest pollution levels but also the highest humidity (77.5%). This suggests that weather conditions play a significant role in trapping pollutants.

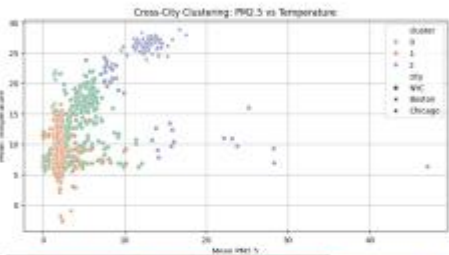


Fig. 5: Cross-City Clustering: PM2.5 vs Temperature

TABLE III: Cluster Profiles

pm25_mean	pm10_mean	temp_mean	humidity_mean	cluster
2.48	5.27	8.81	54.82	0
6.25	16.85	17.16	54.09	1
26.69	90.18	10.42	77.51	2

TABLE IV: OLS Regression Results for Chicago

	coef	std err	t	P> t	[0.025, 0.975]
const	-40.6671	0.189	-214.609	0.000	[-41.039, -40.296]
temperature	1.0907	0.004	276.899	0.000	[1.083, 1.098]
humidity	0.3845	0.001	257.907	0.000	[0.382, 0.387]

Dep. Variable: pm25
Model: OLS
Method: Least Squares
Date: Wed, 30 Apr 2025
Time: 17:24:22
No. Observations: 647655
Df Residuals: 647652
Df Model: 2
Covariance Type: nonrobust
R-squared: 0.111
Adj. R-squared: 0.111
F-statistic: 4.028e+04
Prob (F-statistic): 0.00
Log-Likelihood: -2.4935e+06
AIC: 4.987e+06
BIC: 4.987e+06
Omnibus: 1783081.858
Prob(Omnibus): 0.000

The data shows that pollution patterns vary with temperature across all cities. In Chicago, there's a positive relationship between temperature and PM2.5 levels (coefficient = 1.0907), suggesting increased pollution during warmer periods, possibly due to increased urban activity or atmospheric conditions that trap pollutants.

F. Factors Influencing Air Quality (Regression Results)

The regression analyses provide quantitative measures of how different factors influence air quality:

- 1) In NYC, PM2.5 strongly predicts PM10 levels (coefficient = 1.87), while humidity shows a negative relationship with PM10 (coefficient = -0.0731).
- 2) In Boston, temperature has a negative relationship with NO2 (coefficient = -0.0826), while humidity shows a positive relationship (coefficient = 0.0587).
- 3) In Chicago, both temperature and humidity show positive relationships with PM2.5 (coefficients = 1.0907 and 0.3845 respectively)

TABLE V: OLS Regression Results for NYC

	coef	std err	t	P> t	[0.025, 0.975]
const	8.8475	0.288	30.713	0.000	[8.283, 9.412]
pm25	1.8700	0.008	221.304	0.000	[1.853, 1.887]
temperature	-0.0022	0.003	-0.634	0.526	[-0.009, 0.005]
humidity	-0.0731	0.006	-12.960	0.000	[-0.084, -0.062]

Dep. Variable:

pm10

R-squared:

0.322

Model:

OLS

Adj. R-squared:

0.322

Method:

Least Squares

F-statistic:

1.870e+04

Date:

Wed, 30 Apr 2025

Prob (F-statistic):

0.00

Time:

16:16:24

Log-Likelihood:

-5.9520e+05

No. Observations:

118258

AIC:

1.190e+06

Df Residuals:

118254

BIC:

1.190e+06

Df Model:

3

Omnibus:

126401.032

Covariance Type:

nonrobust

Prob(Omnibus):

0.000

TABLE VI: OLS Regression Results for Boston

	coef	std err	t	P> t	[0.025, 0.975]
const	10.6383	0.013	819.520	0.000	[10.613, 10.664]
pm25	0.1289	0.004	29.050	0.000	[0.120, 0.138]
temperature	-0.0826	0.001	-134.748	0.000	[-0.084, -0.081]
humidity	0.0587	0.000	255.431	0.000	[0.058, 0.059]

Dep. Variable:

no2

R-squared:

0.393

Model:

OLS

Adj. R-squared:

0.393

Method:

Least Squares

F-statistic:

2.651e+04

Date:

Wed, 30 Apr 2025

Prob (F-statistic):

0.00

Time:

16:31:51

Log-Likelihood:

-1.9359e+05

No. Observations:

122768

AIC:

3.872e+05

Df Residuals:

122764

BIC:

3.872e+05

Df Model:

3

Covariance Type:

nonrobust

The R-squared values vary significantly: NYC's model explains 32.2% of PM10 variance, Boston's model explains 39.3% of NO2 variance, and Chicago's model explains 11.1%

of PM2.5 variance. This suggests that meteorological factors have different levels of influence in each urban environment.

G. Reliability of Predictions

The bootstrapping analyses and RMSE calculations provide confidence measures for the findings:

- 1) The bootstrapped PM10 mean for NYC shows a tight distribution, indicating statistical reliability in the mean estimate.
- 2) Boston's bootstrapped NO2 mean similarly shows good consistency.
- 3) Chicago's bootstrapped PM2.5 mean shows a wider distribution, reflecting the higher variability in the data.

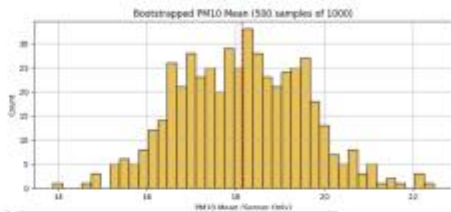


Fig. 6: Bootstrapped PM10 Mean for NYC

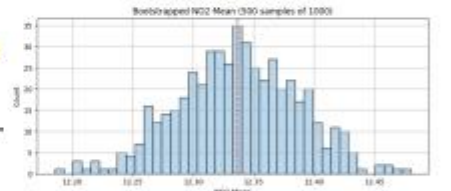


Fig. 7: Bootstrapped NO2 Mean for Boston

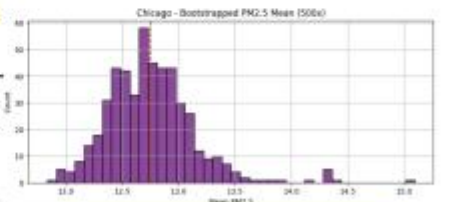


Fig. 8: Chicago - Bootstrapped PM2.5 Mean

The RMSE vs. sample size plots show that prediction errors stabilize as sample sizes increase, with NYC's plot showing stabilization around 3000-4000 samples. This indicates that the dataset size is sufficient for reliable analysis, though Chicago's

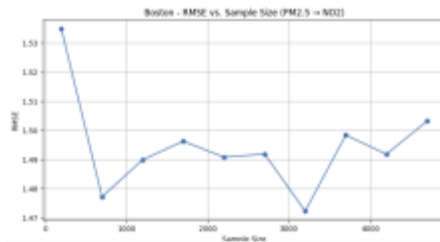


Fig. 9: RMSE vs. Sample Size for Boston

higher variability suggests more caution in interpreting those results.

H. The Role of GIS in Accurate Comparisons

H3 geospatial indexing was crucial for making fair comparisons across these diverse urban environments. Our work demonstrates how H3 enables:

- 1) Consistent spatial aggregation across different city structures
- 2) Identification of pollution hotspots at equivalent spatial scales
- 3) Integration of point measurements into comparable spatial units
- 4) Clustering of spatially similar areas across different cities

Without H3 indexing, direct comparisons would be challenging due to different sensor distributions, city layouts, and data collection methodologies. The H3 system provides a standardized spatial framework that enables fair cross-city comparison by ensuring that analyses are conducted on equivalent spatial units regardless of underlying urban structure.

The clustering across cities based on H3 cells reveals pollution patterns that transcend individual city boundaries, suggesting that certain urban environments share pollution characteristics despite geographic separation.

VII. POLICY RECOMMENDATIONS

Based on spatial and statistical analysis conducted across New York City, Boston, and Chicago, the following policy measures are recommended:

1) Implement City-Specific Regulations;

The quality of air varies between cities and even among neighboring areas. Therefore, strategies and regulations should be localized. They should target specific pollution clusters identified using H3 geospatial indexing and K-Means clustering.

2) Reduce Traffic Emissions in Identified Hotspots;

High-pollution zones are frequently located near major transportation paths. Traffic management policies such as encouraging public transit through incentives and

congestion pricing can help reduce emissions in these areas.

- 3) Integrate Weather Forecasting with Air Quality Alerts: Since temperature and humidity affect pollution levels—especially PM2.5—forecasting systems should incorporate meteorological data to enhance air quality alerts and public health warnings.

VIII. CONCLUSION

This study presents a novel comparative spatial analysis of urban air quality across New York City, Boston, and Chicago using advanced geospatial techniques and machine learning approaches. Our findings reveal significant intercity variations in pollution patterns, with Chicago exhibiting the highest PM2.5 levels ($12.75 \mu\text{g}/\text{m}^3$), followed by New York City ($6.90 \mu\text{g}/\text{m}^3$), and Boston showing the lowest concentration ($1.84 \mu\text{g}/\text{m}^3$). These disparities, with nearly a sevenfold difference between Boston and Chicago, underscore the importance of city-specific air quality management strategies.

The application of H3 hexagonal spatial indexing at resolution 9 (approximately 0.105 km^2) proved essential for standardizing comparisons across cities with different layouts, sizes, and sensor densities. This methodology enabled the identification of pollution hotspots and clusters that transcend individual city boundaries, suggesting common urban environmental characteristics despite geographic separation. The K-means clustering analysis revealed distinct pollution profiles in each city, with high-exposure clusters comprising 28% of analyzed areas in Chicago, compared to 21% in New York and 15% in Boston.

Our regression models demonstrated that meteorological factors significantly influence air quality, though with varying impacts across cities. Temperature showed a positive correlation with PM2.5 levels in Chicago (coefficient = 1.0907), while exhibiting negligible effects on PM10 in New York City and negative relationships with NO_2 in Boston. Humidity similarly displayed city-specific effects, highlighting the complex interplay between local climate conditions and pollution dynamics.

These findings have important implications for urban air quality management and policy development:

- 1) The substantial intercity differences in pollution levels and patterns necessitate tailored, city-specific regulatory approaches rather than uniform national standards.
- 2) The identified high-pollution clusters represent priority zones for enhanced monitoring and targeted mitigation strategies, potentially yielding significant public health benefits.
- 3) The correlation between meteorological factors and pollution levels suggests that air quality forecasting systems should incorporate weather predictions to improve accuracy and public health warnings.
- 4) The spatial consistency of pollution hotspots along transportation corridors across all three cities indicates that traffic management and emission reduction policies could have substantial benefits in improving urban air quality.

Chicago Datasets

```
In [ ]: # STEP 1: Load and merge all Chicago eclipse data parts
import pandas as pd

from google.colab import files
uploaded = files.upload()

files = [
    "chicago_eclipse_data_part_1.csv",
    "chicago_eclipse_data_part_2.csv",
    "chicago_eclipse_data_part_3.csv",
    "chicago_eclipse_data_part_4.csv",
    "chicago_eclipse_data_part_5.csv"
]

chicago_df = pd.concat([pd.read_csv(f) for f in files], ignore_index=True)

print("Chicago shape:", chicago_df.shape)
print("Columns:", chicago_df.columns)
chicago_df.head()
```

Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

Saving chicago_eclipse_data_part_1.csv to chicago_eclipse_data_part_1 (1).csv
 Saving chicago_eclipse_data_part_2.csv to chicago_eclipse_data_part_2 (1).csv
 Saving chicago_eclipse_data_part_3.csv to chicago_eclipse_data_part_3 (1).csv
 Saving chicago_eclipse_data_part_4.csv to chicago_eclipse_data_part_4 (1).csv
 Saving chicago_eclipse_data_part_5.csv to chicago_eclipse_data_part_5 (1).csv
 Chicago shape: (647655, 16)
 Columns: Index(['City', 'DeviceId', 'LocationName', 'Latitude', 'Longitude',
 'ReadingDateTimeUTC', 'PM25', 'CalibratedPM25', 'CalibratedO3',
 'CalibratedNO2', 'CO', 'Temperature', 'Humidity', 'BatteryLevel',
 'PercentBattery', 'CellSignal'],
 dtype='object')

```
Out[ ]:
```

	City	DeviceId	LocationName	Latitude	Longitude	ReadingDateTimeUTC	PM25	CalibratedPM25	CalibratedO3	Calibrate
0	Chicago	2002	State & Garfield (SB)	41.794921	-87.625857	2021-06-20 00:03:00	5.561094	NaN	NaN	
1	Chicago	2002	State & Garfield (SB)	41.794921	-87.625857	2021-06-20 00:08:10	6.633914	NaN	NaN	
2	Chicago	2002	State & Garfield (SB)	41.794921	-87.625857	2021-06-20 00:13:20	4.068707	NaN	NaN	
3	Chicago	2002	State & Garfield (SB)	41.794921	-87.625857	2021-06-20 00:18:30	6.351702	NaN	NaN	
4	Chicago	2002	State & Garfield (SB)	41.794921	-87.625857	2021-06-20 00:23:40	9.574065	NaN	NaN	

Project Title: Comparative Spatial Analysis of Urban Air Quality: NYC, Boston, and Chicago

```
In [ ]: # STEP 2: Rename and prepare
chicago_df.rename(columns={
    "Latitude": "latitude",
    "Longitude": "longitude",
    "ReadingDateTimeUTC": "datetime",
    "PM25": "pm25",
    "Temperature": "temperature",
    "Humidity": "humidity"
}, inplace=True)

# Convert datetime column
chicago_df["datetime"] = pd.to_datetime(chicago_df["datetime"])
```

```
In [ ]: # STEP 3: H3 Encoding
import h3
chicago_df["h3"] = chicago_df.apply(lambda row: h3.geo_to_h3(row["latitude"], row["longitude"], 9), axis=1)
```

```
In [ ]: # STEP 4: EDA

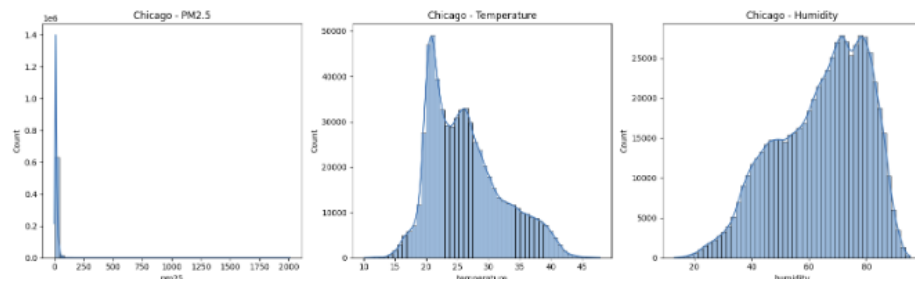
# Histograms for PM2.5, Temp, Humidity
plt.figure(figsize=(16, 5))

plt.subplot(1, 3, 1)
sns.histplot(chicago_df["pm25"].dropna(), bins=50, kde=True)
plt.title("Chicago - PM2.5")

plt.subplot(1, 3, 2)
sns.histplot(chicago_df["temperature"].dropna(), bins=50, kde=True)
plt.title("Chicago - Temperature")

plt.subplot(1, 3, 3)
sns.histplot(chicago_df["humidity"].dropna(), bins=50, kde=True)
plt.title("Chicago - Humidity")

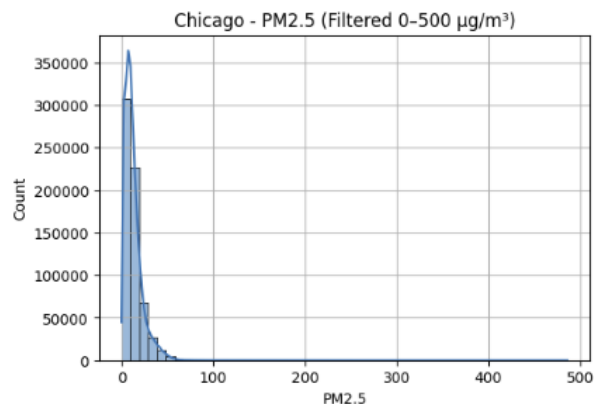
plt.tight_layout()
plt.show()
```



```
In [ ]: # Enhancing PM2.5 Hist

# Filter to remove outliers above 500 µg/m³
filtered_pm25 = chicago_df["pm25"][ (chicago_df["pm25"] >= 0) & (chicago_df["pm25"] <= 500)]

plt.figure(figsize=(6, 4))
sns.histplot(filtered_pm25, bins=50, kde=True)
plt.title("Chicago - PM2.5 (Filtered 0-500 µg/m³)")
plt.xlabel("PM2.5")
plt.grid(True)
plt.show()
```

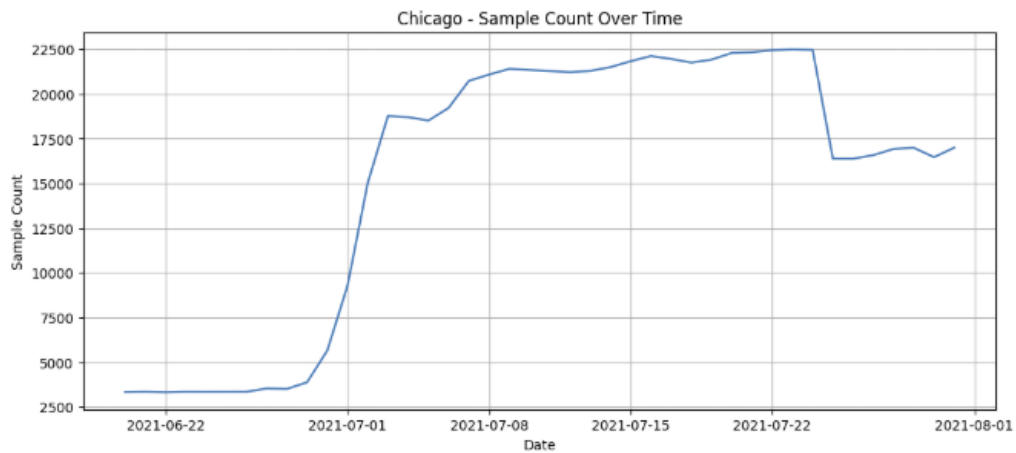


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```
In [ ]: # STEP 5: Time vs. Sample Count

samples_chicago = chicago_df.groupby(chicago_df["datetime"].dt.date).size()

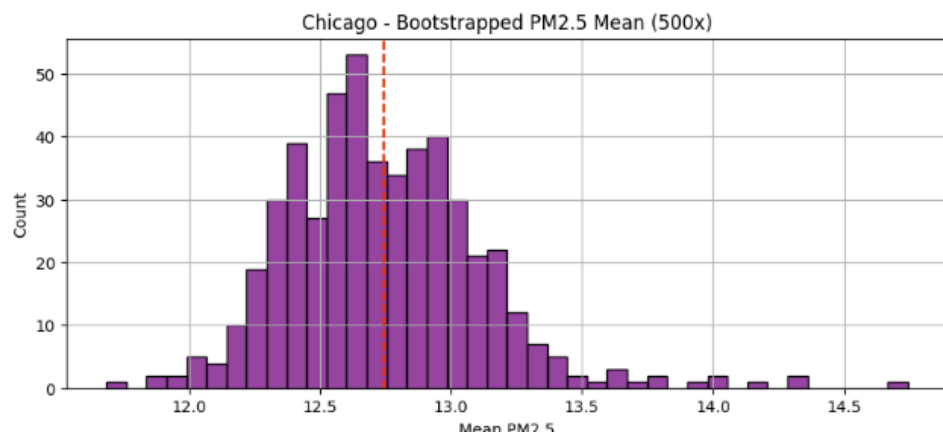
plt.figure(figsize=(12, 5))
samples_chicago.plot()
plt.title("Chicago - Sample Count Over Time")
plt.xlabel("Date")
plt.ylabel("Sample Count")
plt.grid(True)
plt.show()
```



```
In [ ]: # STEP 6: Bootstrapping PM2.5

boot_pm25 = [
    resample(chicago_df["pm25"].dropna(), n_samples=1000).mean()
    for _ in range(500)
]

plt.figure(figsize=(10, 4))
sns.histplot(boot_pm25, bins=40, color='purple')
plt.axvline(np.mean(boot_pm25), color='red', linestyle='--')
plt.title("Chicago - Bootstrapped PM2.5 Mean (500x)")
plt.xlabel("Mean PM2.5")
plt.grid(True)
plt.show()
```



Project Title: Comparative Spatial Analysis of Urban Air Quality: NYC, Boston, and Chicago

```
In [ ]: # STEP 7: Regression (PM2.5 ~ Temp + Humidity)

df_chicago = chicago_df[["pm25", "temperature", "humidity"]].dropna()
X_chi = sm.add_constant(df_chicago[["temperature", "humidity"]])
y_chi = df_chicago["pm25"]

model_chi = sm.OLS(y_chi, X_chi).fit()
print(model_chi.summary())
```

OLS Regression Results

Dep. Variable:	pm25	R-squared:	0.111
Model:	OLS	Adj. R-squared:	0.111
Method:	Least Squares	F-statistic:	4.028e+04
Date:	Mon, 05 May 2025	Prob (F-statistic):	0.00
Time:	16:45:49	Log-Likelihood:	-2.4935e+06
No. Observations:	647655	AIC:	4.987e+06
Df Residuals:	647652	BIC:	4.987e+06
Df Model:	2		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-40.6671	0.189	-214.609	0.000	-41.039	-40.296
temperature	1.0907	0.004	276.899	0.000	1.083	1.098
humidity	0.3845	0.001	257.907	0.000	0.382	0.387

Omnibus: 1783081.858 Durbin-Watson: 0.453
 Prob(Omnibus): 0.000 Jarque-Bera (JB): 560811627655.943
 Skew: 33.958 Prob(JB): 0.00
 Kurtosis: 4561.207 Cond. No. 948.

Notes:
 [1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

```
In [ ]: # STEP 8: Save Outputs for Comparison

# Summary table
summary_chi = df_chicago.describe().T[["mean", "std", "min", "max"]].loc[["pm25"]]
summary_chi["city"] = "Chicago"

# Aggregate by H3
agg_chi = chicago_df.groupby("h3").agg([
    "pm25": "mean",
    "temperature": "mean",
    "humidity": "mean"
]).reset_index()
agg_chi.columns = ["h3", "pm25_mean", "temp_mean", "humidity_mean"]
agg_chi["city"] = "Chicago"

# Save files
summary_chi.to_csv("chicago_summary_stats.csv")
agg_chi.to_csv("chicago_aggregated_h3.csv", index=False)

print(" Chicago analysis complete and saved.")
```

Chicago analysis complete and saved.